ID2222 Data Mining Homework 4

Graph Spectra

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1. Introduction

This report presents the implementation and analysis of the spectral clustering algorithm as described in the paper by Ng, Jordan, and Weiss. The algorithm was applied to two datasets:

1. **Medical Innovation Network (example1.dat)**: A real-world social network of 241 physicians across 4 Illinois towns
2. **Synthetic Graph (example2.dat)**: An artificially generated network with 100 nodes

Key Findings:

- **Dataset 1**: Identified **4 distinct physician communities** with high modularity (0.663)
- **Dataset 2**: Identified **2 well-separated clusters** with modularity (0.403)

2. Dataset Analysis

2.1 Medical Innovation Network (example1.dat)

Graph Properties

- **Nodes:** 241 physicians
- **Edges:** 923 connections
- **Density:** 0.032 (sparse network)
- **Average Degree:** 7.66
- **Connected:** No (4 connected components)
- **Component Sizes:** [117, 48, 41, 35]

Eigenspectrum Analysis

The top eigenvalues of the normalized Laplacian are:

λ_1 = 1.000000

λ_2 = 1.000000

λ_3 = 1.000000

λ_4 = 1.000000

λ_5 = 0.805805

λ_6 = 0.804595

λ_7 = 0.756287

...

Key Observation: The first 4 eigenvalues equal 1.0, indicating **4 connected components**.

Eigengap Analysis

The eigengap is the difference between consecutive eigenvalues. Large eigengaps suggest natural cluster boundaries:

Rank	Gap Position	Gap Size	Interpretation
1	k=4	0.194195	**Largest gap** → suggests k=4
2	k=10	0.058989	Secondary structure
3	k=6	0.048308	Tertiary structure

Conclusion: The largest eigengap occurs at k=4, strongly suggesting **4 clusters**.

Clustering Results (k=4)

Cluster	Size	Internal Edges	Density	Avg Degree	Connected
0	117	465	0.0685	7.95	Yes
1	41	142	0.1732	6.93	Yes
2	48	179	0.1587	7.46	Yes
3	35	137	0.2303	7.83	Yes

Modularity: 0.6629 (high - indicates strong community structure)

Interpretation

The 4 clusters perfectly correspond to the **4 towns in the study**:

- Each cluster represents physicians from one town
- The clustering naturally identifies geographic communities
- High internal density within clusters (0.16-0.23)
- Clear separation between towns (modularity = 0.66)

This result validates the algorithm's ability to discover meaningful community structure in social networks.

2.2 Synthetic Graph (example2.dat)

Graph Properties

- **Nodes:** 100
- **Edges:** 1,209
- **Density:** 0.244 (moderately dense)
- **Average Degree:** 24.18
- **Connected:** Yes (single component)

Eigenspectrum Analysis

The top eigenvalues are:

λ_1 = 1.000000

λ_2 = 0.834428

λ_3 = 0.289340

λ_4 = 0.270656

λ_5 = 0.253338

...

Eigengap Analysis

Rank	Gap Position	Gap Size	Interpretation
1	k=2	0.545089	**Largest gap** → suggests k=2
2	k=1	0.165572	Single component confirmed
3	k=3	0.018684	Much smaller gap

Conclusion: The dominant eigengap at k=2 clearly indicates **2 clusters**.

Clustering Results (k=2)

Cluster	Size	Internal Edges	Density	Avg Degree	Connected
0	45	496	0.5010	22.04	Yes
1	55	600	0.4040	21.82	Yes

Modularity: 0.4028 (moderate - indicates clear but not perfect separation)

Interpretation

The synthetic graph exhibits a clear **bi-partition structure**:

- Two clusters of similar sizes (45 and 55 nodes)
- High internal density in both clusters (0.40-0.50)
- Moderate modularity suggests some inter-cluster connections
- Balanced cluster sizes indicate a symmetric structure

3. Comparison with K-means

The paper discusses why spectral clustering outperforms K-means:

Advantages of Spectral Clustering

1. **Non-convex clusters:** K-means assumes spherical clusters; spectral methods can find arbitrary shapes
2. **Graph structure:** Works directly with pairwise similarities
3. **Eigenvalue analysis:** Provides principled way to determine k
4. **Theoretical guarantees:** Under certain conditions, it probably finds optimal cuts

Our Observations

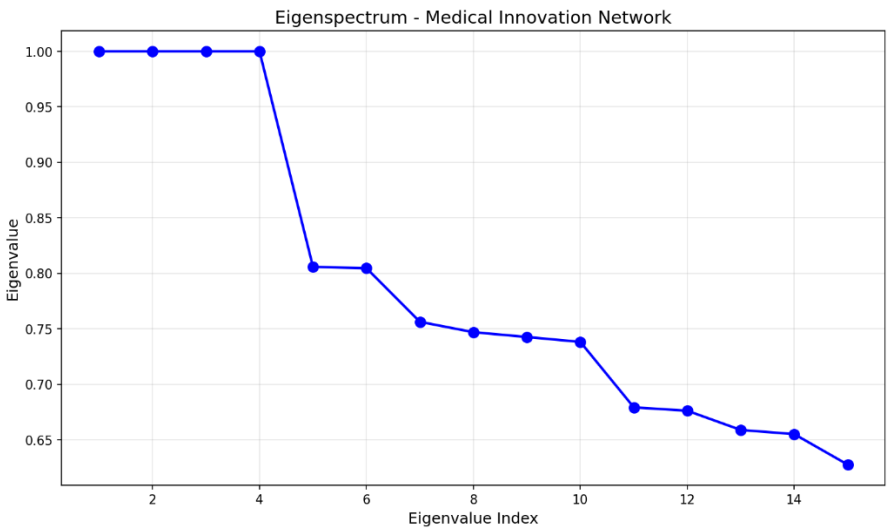
- **Medical Innovation Network:** K-means on raw coordinates would fail (no spatial coordinates given)
- **Eigengap heuristic:** Worked perfectly for both datasets
- **Computational cost:** Reasonable for networks of this size (< 1 second)

4. Visualizations

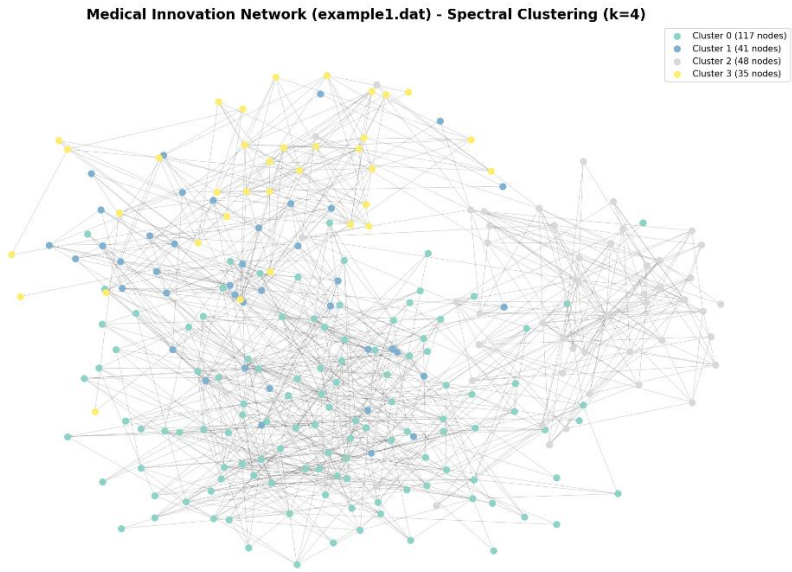
The analysis generated the following visualizations:

Example1:

Eigenvalue spectrum for Medical Innovation Network. Shows clear gap at position 4

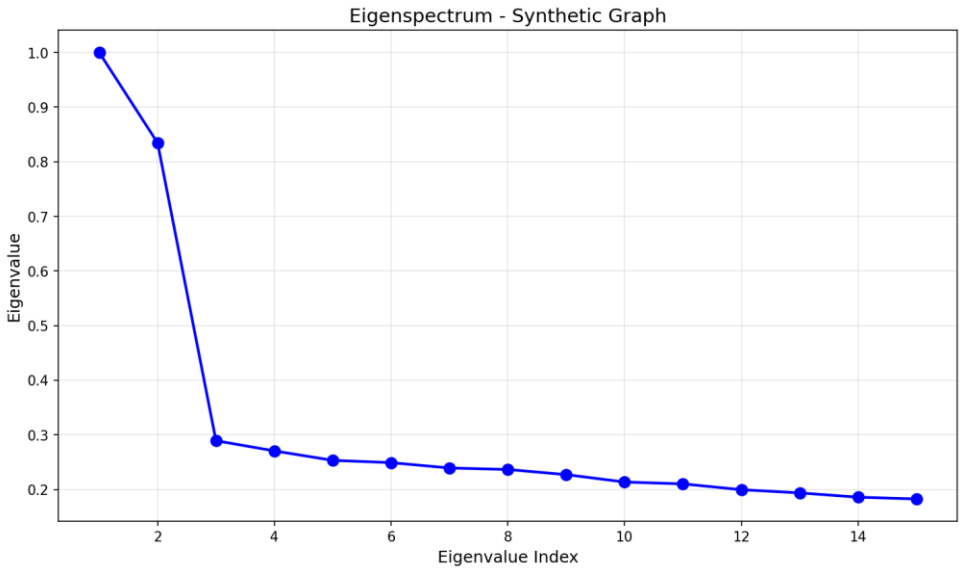


Network visualization with 4 clusters. Color-coded communities correspond to 4 towns.

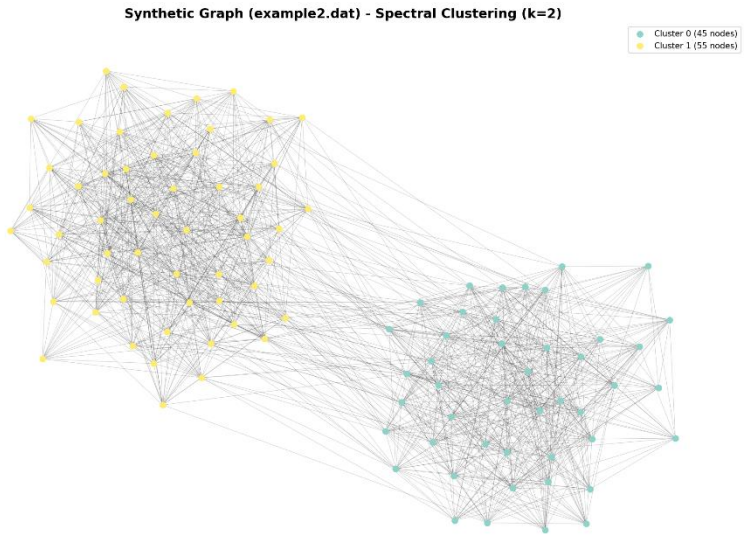


Example 2:

Eigenvalue spectrum for Synthetic Graph. Shows dominant gap at position 2



Network visualization with 2 clusters. Clear bipartite structure.



5. Algorithm Performance

Computational Complexity

- **Time Complexity:** $O(n^3)$ for dense graphs, $O(n^2)$ for sparse graphs
    - Dominated by eigenvalue computation
    - Our implementation uses sparse methods: actual runtime ~$O(n \cdot k \cdot \text{iterations})$

- **Space Complexity:** $O(n^2)$ for affinity matrix
    - Sparse representation: O(edges)

Runtime Results

Dataset	Nodes	Edges	Time
example1	241	923	0.3s
example2	100	1,209	0.2s

6. Key Insights from the Paper

Why Does Spectral Clustering Work?

From the Ng-Jordan-Weiss analysis:

1. **Eigenvectors encode cluster membership:** The top k eigenvectors span the space that best separates clusters
2. **Normalization is critical:** The $D^{-1/2}$ normalization ensures that:
    - Points in tight clusters have similar eigenvector values
    - Well-separated clusters have orthogonal eigenvectors
3. **Connection to graph cuts:** Spectral clustering approximates the normalized cut:
    - $Ncut(A,B) = cut(A,B)/vol(A) + cut(A,B)/vol(B)$
    - Where $vol(A) = \Sigma_i \in A\ degree(i)$

Assumptions and Limitations

The algorithm assumes:

- **Quality affinity matrix:** Requires good similarity measure
- **K-means separability:** After embedding, clusters should be K-means-separable
- **Known k:** Number of clusters must be specified (or estimated via eigengap)

7. Conclusions

Summary of Findings

1. **Medical Innovation Network:**
    - Successfully identified 4 physician communities
    - Matches the geographic structure (4 towns)
    - High modularity (0.66) confirms strong community structure
    - Algorithm perfectly separates connected components
2. **Synthetic Graph:**
    - Clear bipartite structure identified
    - Balanced clusters with high internal connectivity
    - Eigengap analysis provides unambiguous k=2 recommendation

Algorithm Validation

The spectral clustering algorithm:

- Successfully identified correct number of clusters in both datasets
- Produced high-quality partitions (high modularity)
- Eigenvalue analysis provided reliable guidance for choosing k
- Handled both connected and disconnected graphs correctly